Pre-Class Assignment: Chapter 10

Pricing with Exogenous Judgment

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

Reading: Gans, Chapter 10, Sections 10.1 (all subsections), 10.2.1 (Monopolist Supply), 10.2.3 (Versioning), and the chapter Conclusion. Sections 10.2.2 and 10.2.4 are optional.

Instructions: Answer each question in 1–3 sentences. These questions are designed to be straightforward if you did the reading.

You must submit this assignment in person. The assignment must be handwritten. No photocopies will be accepted.

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1. Gans introduces a parameter α. In your own words, what does α measure, and why does it matter?

2. A customer is called a **1-default agent** if $\alpha < p$ and a **2-default agent** if $\alpha > p$. What does it mean, behaviorally, to be a 1-default versus a 2-default? Give a one-sentence example of each kind of customer.

3. What is the willingness to pay of a 1-default for a perfect prediction, and why does $(1 - p)$ appear in the answer?

4. The willingness-to-pay diagram in Figure 10.2 is a **triangle**, peaked at $\alpha = p$. In one sentence, explain why the customer with $\alpha = p$ has the *highest* willingness to pay.

6. In Section 10.2.1, Gans shows that when $p > 1/2$, the monopolist's optimal product is the corner $\{e_1, e_2\} = \{1, \Lambda - 1\}$. What does this product look like (in terms of which state it identifies perfectly and which it identifies noisily), and why does the monopolist choose this corner rather than the symmetric corner $\{\Lambda - 1, 1\}$?

7. Figure 10.3 shows the monopoly outcome on the WTP triangle. Which customers end up *buying* the AI prediction at price $\hat{P}_m$, and which *two* groups of customers end up excluded from the market?

8. In Section 10.2.3, Gans considers a monopolist offering both a perfect prediction $\{1, 1\}$ at price $P_{1,1}$ and a degraded prediction $\{e_1, 1\}$ at a lower price $P_{e_1,1}$ — the standard "Lite vs. Pro" versioning strategy used in software markets. What is the surprising equilibrium outcome about the *quantity* of the Lite version sold?

9. Why does versioning AI predictions fail to deliver the kind of surplus extraction that versioning software (e.g., Word vs. Word Lite) delivers? Answer in one or two sentences referring to where the *value of a prediction* is concentrated.

10. In one sentence, state the main takeaway of Chapter 10.